

# PROJECT PRESENTATION

## Feature Extraction & Price Prediction for Mobile Phones

*Using Feature Extraction and Machine Learning Techniques*

### Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Google Colab

### Submitted By

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### PROJECT FOCUS

- ❖ Mobile phone price prediction
- ❖ Feature extraction & engineering
- ❖ EDA, skewness & outlier analysis
- ❖ Machine learning model comparison
- ❖ Feature importance & business insights

# Business Problem & Objective

- ❖ Mobile-phone prices depend on multiple specifications including memory, RAM, battery, cameras, processor and product attributes.
- ❖ **Objective:** build a predictive model that estimates mobile-phone price from available features.
- ❖ Perform feature extraction to identify the most influential predictive features.
- ❖ **Business value:** data-driven pricing, product positioning and marketing decisions.

# End-to-End Project Workflow



- ❖ The workflow combines statistical analysis with machine learning so that feature selection is supported by both EDA and predictive performance.

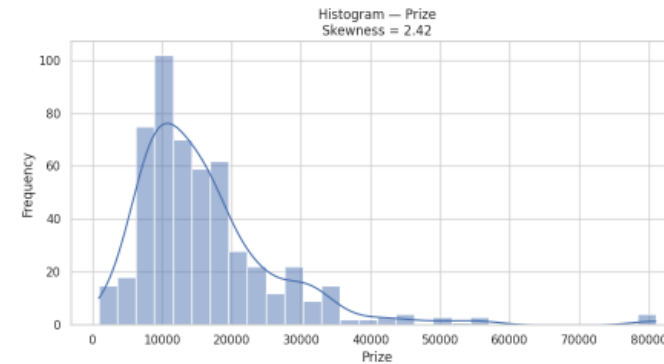
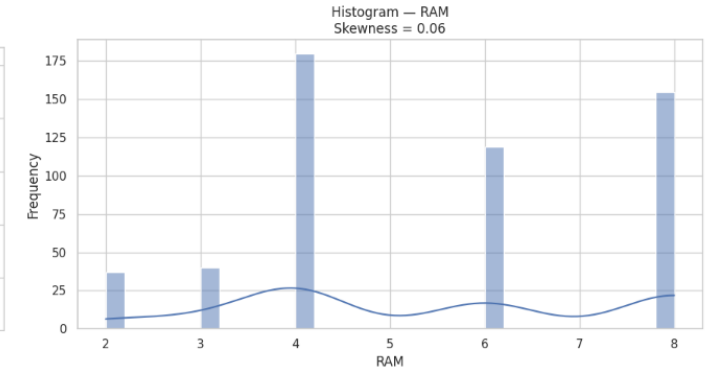
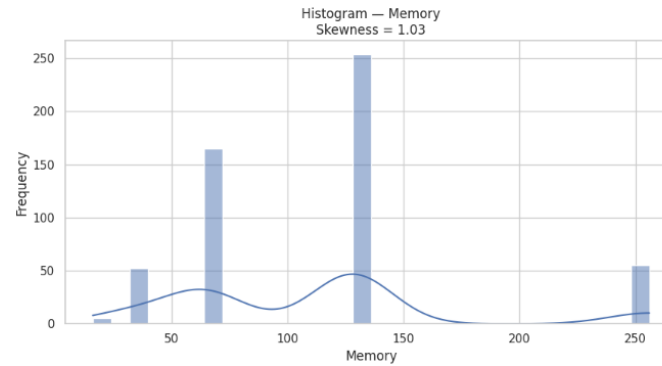
# Dataset Overview & Data Quality

| Metric             | Value   |
|--------------------|---------|
| Rows               | 531     |
| Features           | 11      |
| Target             | Price   |
| Duplicates handled | Removed |
| Missing values     | Handled |

- ❖ Exact duplicate records are removed.
- ❖ Target price is converted to numeric and invalid target rows are excluded.
- ❖ Numerical missing values use median imputation.
- ❖ Categorical missing values use most-frequent imputation.
- ❖ Raw specification text is converted into machine-learning features.

# EDA — Univariate Analysis, Histogram & Skewness

| Feature       | Skewness |
|---------------|----------|
| AI Lens       | 3.57     |
| Price         | 2.42     |
| Mobile Height | 1.42     |
| Memory        | 1.03     |
| RAM           | 0.06     |
| Battery_      | -3.33    |



- ❖ Histograms and KDE curves show the distribution of numerical features.
- ❖ Positive skewness indicates a longer right tail; negative skewness indicates a longer left tail.
- ❖ Boxplots help identify extreme observations.
- ❖ Interpret skewness using the actual values from the notebook rather than assuming a distribution is normal.

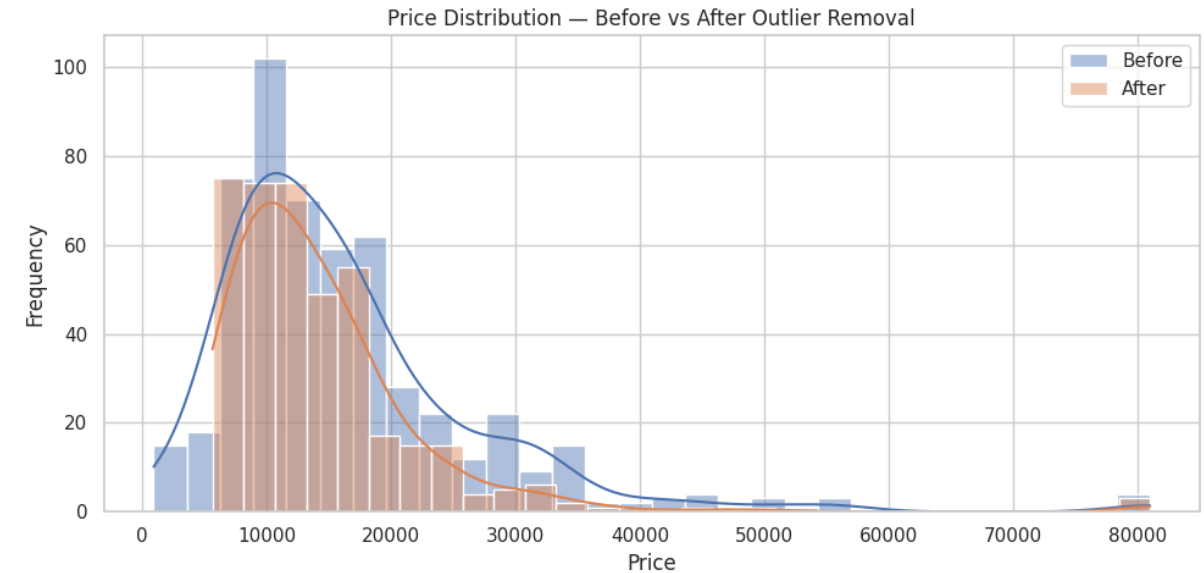
# Target Variable — Price Distribution

| Statistic | Value   |
|-----------|---------|
| Minimum   | ₹920    |
| Median    | ₹13,999 |
| Mean      | ₹16,306 |
| Maximum   | ₹80,999 |
| Skewness  | 2.42    |

- ❖ Mean price is approximately ₹16,306; median is ₹13,999.
- ❖ Price skewness is 2.42, showing a long upper tail.
- ❖ Premium phones can be genuine high-value observations and should not be deleted solely because of statistical outlier rules.

# Outlier Detection & Removal — IQR Method

| Feature       | Outlier Count |
|---------------|---------------|
| Memory        | 55            |
| Mobile Height | 32            |
| Price         | 25            |
| RAM           | 0             |
| AI Lens       | 0             |
| Battery_      | 0             |



- ❖ IQR bounds: Lower =  $Q1 - 1.5 \times IQR$ ; Upper =  $Q3 + 1.5 \times IQR$ .
- ❖ Predictor outliers are removed to reduce distortion from extreme measurements.
- ❖ Price outliers are reviewed separately because premium devices may be legitimate.
- ❖ Model performance is finally judged on unseen test data.

# Feature Extraction & Engineering

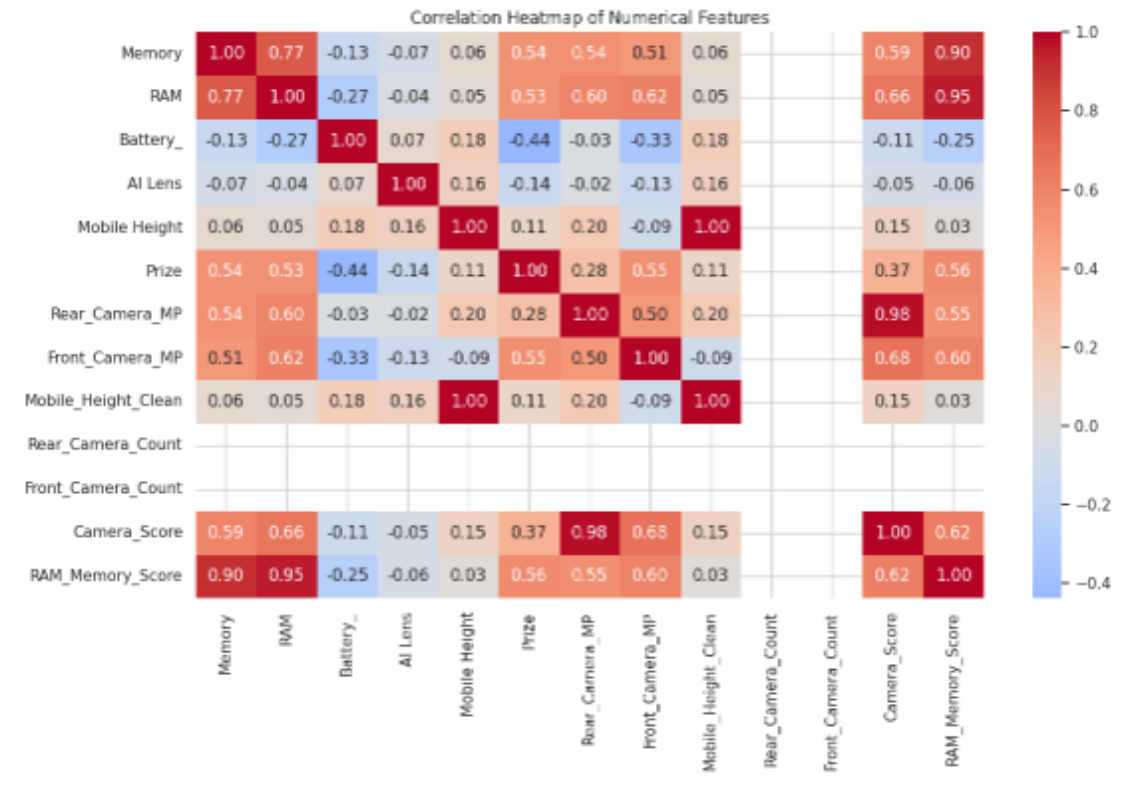
| Raw information | Extracted feature   | Purpose              |
|-----------------|---------------------|----------------------|
| Rear Camera     | Rear_Camera_MP      | Numeric camera MP    |
| Front Camera    | Front_Camera_MP     | Numeric camera MP    |
| Mobile Height   | Mobile_Height_Clean | Clean numeric height |
| RAM + Memory    | RAM_Memory_Score    | Interaction feature  |

- ❖ Camera specifications are transformed from text into numeric features.
- ❖ Camera count and Camera\_Score can further represent imaging capability.
- ❖ Feature engineering makes raw specifications more usable for regression models.
- ❖ Example : **Rear Camera:** "50MP + 8MP + 2MP" into Rear\_Camera\_MP = 50



# Correlation Analysis

| Feature       | Correlation |
|---------------|-------------|
| Memory        | 0.564       |
| RAM           | 0.529       |
| Mobile Height | 0.168       |
| AI Lens       | -0.156      |
| Battery_      | -0.046      |



- ❖ Correlation is an initial feature-screening technique.
- ❖ A value close to +1 or -1 indicates a stronger linear association.
- ❖ Correlation does not establish causation.
- ❖ Non-linear relationships are better captured by tree-based models, so final importance is validated using permutation importance.

# Preprocessing & One-Hot Encoding

- ❖ Numerical features → median imputation.
- ❖ Categorical features → most-frequent imputation + One-Hot Encoding.
- ❖ One-Hot Encoding converts categories such as Model, Colour and Processor into binary numerical features.
- ❖ `handle_unknown='ignore'` allows new/unseen categories during testing and future prediction.
- ❖ `ColumnTransformer` + `Pipeline` keeps preprocessing consistent between training and testing.

# Machine Learning Models

- ❖ Five regression algorithms are trained on the same processed data.
- ❖ Ensemble methods are expected to perform better when price relationships are non-linear.
- ❖ Model selection is based on test performance and cross-validation stability.

| Model             | Purpose             |
|-------------------|---------------------|
| Linear Regression | Baseline            |
| Decision Tree     | Non-linear rules    |
| Random Forest     | Bagging ensemble    |
| Extra Trees       | Randomized ensemble |
| Gradient Boosting | Boosting ensemble   |

# Model Evaluation & Comparison

| Model             | MAE           | RMSE   | R <sup>2</sup> |
|-------------------|---------------|--------|----------------|
| Linear Regression | ₹1,341        | ₹2,385 | 0.8559         |
| Extra Trees       | <b>₹1,074</b> | ₹2,583 | 0.8309         |
| Gradient Boosting | ₹1,354        | ₹2,808 | 0.8002         |
| Random Forest     | ₹1,346        | ₹3,063 | 0.7622         |
| Decision Tree     | ₹1,676        | ₹3,865 | 0.6215         |

- ❖ Lower MAE and RMSE are better; higher R<sup>2</sup> is better.
- ❖ Linear Regression has **RMSE ₹2,385 and R<sup>2</sup> 0.8559**, while Extra Trees has the **lowest MAE of ₹1,074**.

# Best Model — Linear Regression

**₹1,341**

MAE

**₹2,385**

RMSE

**0.8559**

Test  $R^2$

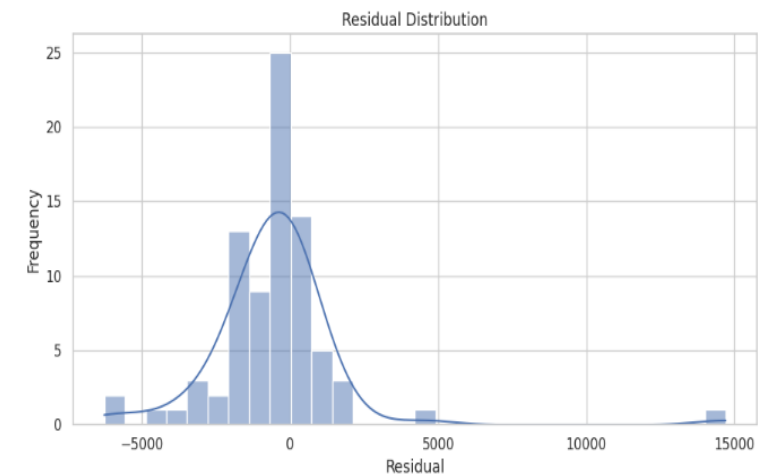
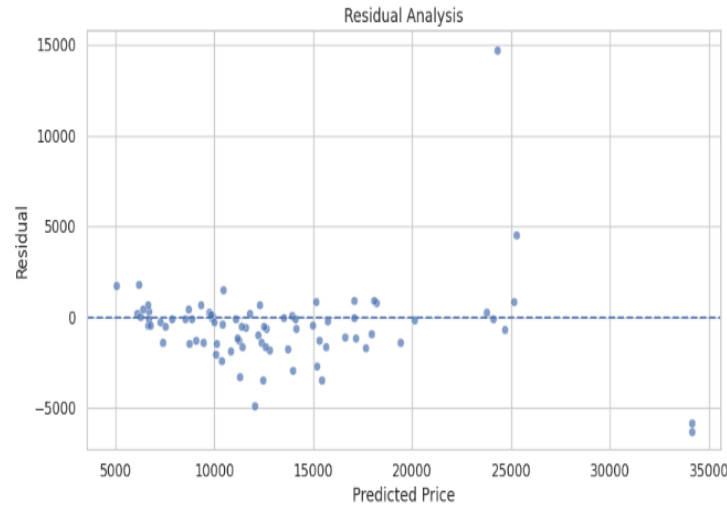
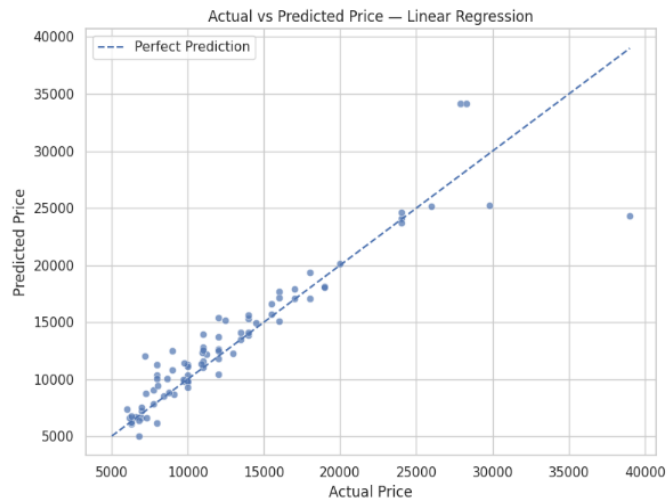
**5-Fold**

Validation

- ❖ Linear Regression explains approximately 85.59% of the variation in mobile-phone prices on the test set.
- ❖ The final notebook also checks train-vs-test  $R^2$  to identify overfitting.
- ❖ Cross-validation is used to assess whether the performance is stable across folds.

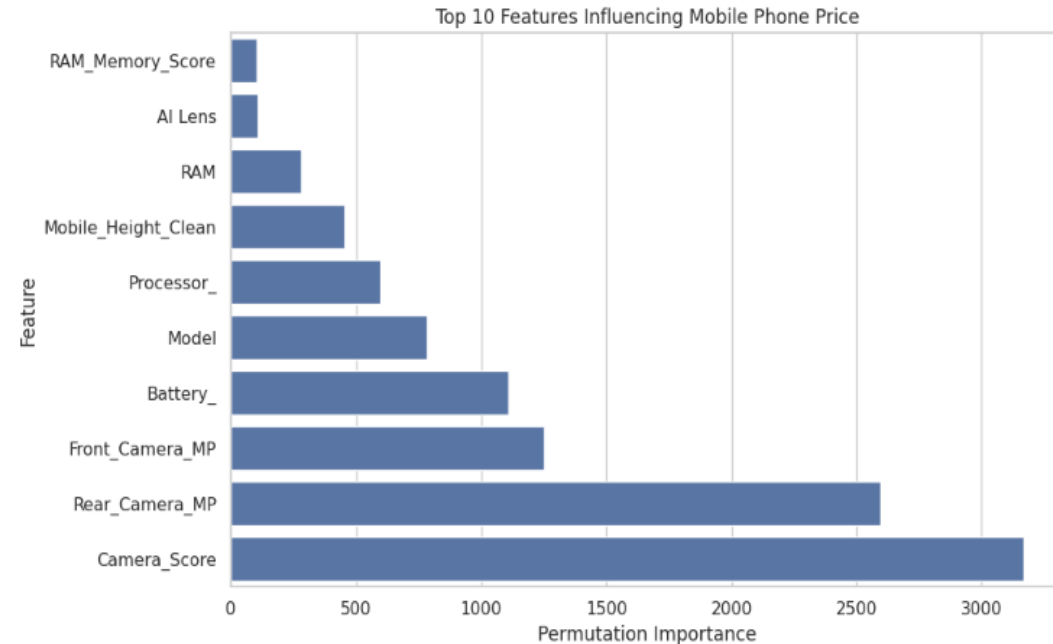
# Actual vs Predicted & Residual Analysis

- ❖ Actual-vs-predicted points should be close to the 45° perfect-prediction line.
- ❖ Residuals should be distributed around zero without strong systematic patterns.
- ❖ Large residuals can indicate unusual products or missing variables.
- ❖ Residual analysis helps validate whether the model is systematically under- or over-predicting.



# Feature Importance Analysis

- ❖ Permutation importance identified the most influential features affecting mobile-phone price. The ranking is based on performance degradation after shuffling each feature on the unseen test set.
- ❖ A larger importance value means shuffling that feature causes a larger deterioration in model performance.
- ❖ Permutation importance identifies Camera\_Score and Rear\_Camera\_MP as the strongest predictive features in the final model.
- ❖ Importance indicates predictive usefulness, not direct causal impact on price.



**Top 10**

Features

**Permutation**

Method

**Test Set**

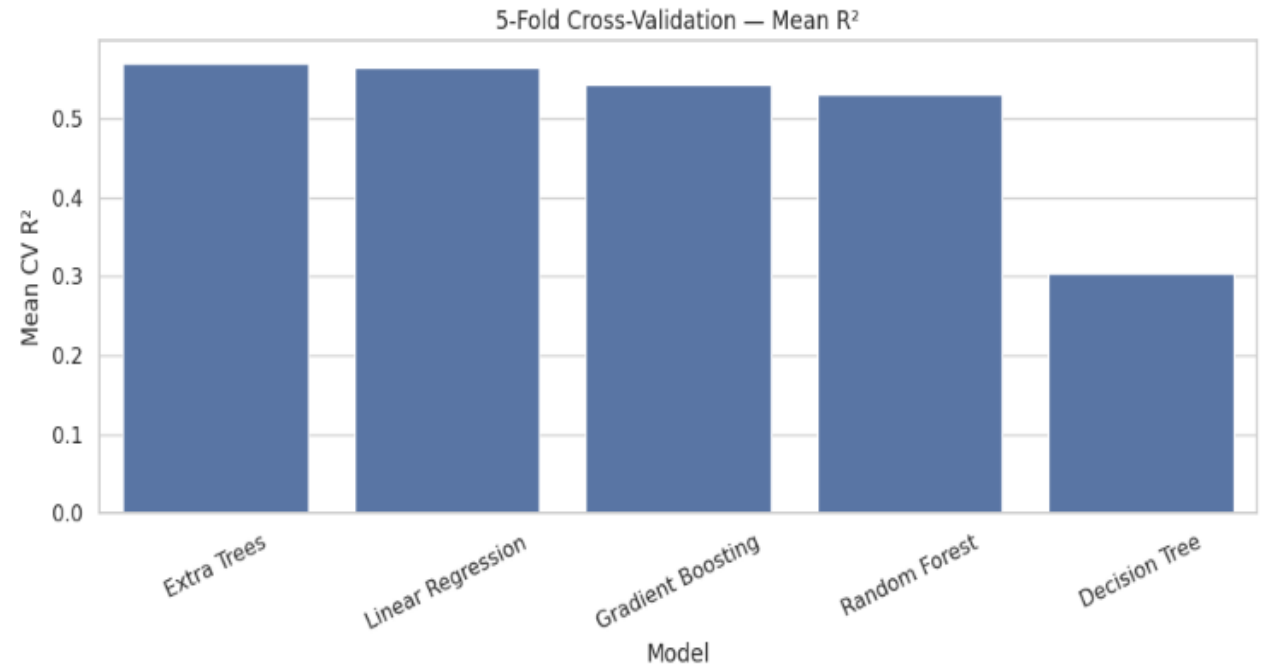
Validation

**R<sup>2</sup>**

Fit

# 5-Fold Cross-Validation & Overfitting Control

- ❖ Five-fold cross-validation evaluates model performance over multiple train/validation splits.
- ❖ A small train-test  $R^2$  gap is preferable to a very high training score with weak test performance.
- ❖ Tree depth, minimum samples, learning rate and ensemble averaging help control overfitting.
- ❖ The goal is not to maximize training  $R^2$ , but to maximize reliable unseen-data performance.
- ❖ Linear Regression was selected based on the final held-out test-set performance, where it achieved the highest  $R^2$  and lowest RMSE; cross-validation was used as a stability check.





# Business Recommendations

- ❖ Use the model as a pricing benchmark rather than the sole pricing decision.
- ❖ Focus pricing analysis on features that rank highest in the final feature-importance results.
- ❖ Separate genuine premium devices from data errors when handling outliers.
- ❖ Combine model predictions with competitor prices, brand strength, launch timing, discounts and demand.
- ❖ Retrain periodically as new phones and prices enter the market.
- ❖ Add future variables such as brand, display technology, refresh rate, chipset benchmark, 5G support and launch date.

# Final Conclusion

- ❖ Raw mobile specifications were transformed into a structured predictive dataset.
- ❖ EDA covered distributions, skewness, categorical patterns, correlations and outliers.
- ❖ Feature extraction converted camera and dimension specifications into useful numeric features.
- ❖ Five regression models were compared using MAE, RMSE and  $R^2$ .
- ❖ Linear Regression achieved a test  $R^2$  of 0.8559, explaining approximately 85.59% of the variation in mobile-phone prices.
- ❖ The solution can support data-driven pricing, product positioning and marketing decisions.

# References & Tools

- ❖ Python — programming and data analysis
- ❖ pandas — data manipulation
- ❖ NumPy — numerical computation
- ❖ Matplotlib & Seaborn — visualization
- ❖ scikit-learn — preprocessing, regression, metrics and cross-validation
- ❖ Project notebook — Mobile\_Phone\_Price\_Prediction\_COMPLETE.ipynb
- ❖ Dataset — mobile\_phone\_dataset.csv

**Thank You**